IE 203 PS 7 – Solutions

Q1. Newton's Method for One-Dimensional Optimization

Solution

Newton's Method

Newton's method is based on exploiting the quadratic approximation of the function θ at a given point λ_k. This quadratic approximation q is given by

$$q(\lambda) = \theta(\lambda_k) + \theta'(\lambda_k)(\lambda - \lambda_k) + \frac{1}{2}\theta''(\lambda_k)(\lambda - \lambda_k)^2.$$

The point λ_{k+1} is taken to be the point where the derivative of q is equal to zero. This yields $\theta'(\lambda_k) + \theta''(\lambda_k)(\lambda_{k+1} - \lambda_k) = 0$, so that

$$\lambda_{k+1} = \lambda_k - \frac{\theta'(\lambda_k)}{\theta''(\lambda_k)}. \tag{8.8}$$

Reference: M. S. Bazaraa, H. D. Sherali, C. M. Shetty, Nonlinear Programming: Theory and Algorithms, 3rd Edition, 2006

The first and second derivatives of the function are:

$$f'(x) = 4x^3 - 12x^2 + 8x$$
$$f''(x) = 12x^2 - 24x + 8$$

The Newton's method update rule is given by:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

We start with $x_1 = 3.0000$ and iterate until $|x_{k+1} - x_k| < 0.1$.

Iteration 1

$$\begin{aligned} x_1 &= 3 \\ f'(3) &= 4(3)^3 - 12(3)^2 + 8(3) = 108 - 108 + 24 = 24 \\ f''(3) &= 12(3)^2 - 24(3) + 8 = 108 - 72 + 8 = 44 \\ x_2 &= 3 - \frac{24}{44} \approx 2.4545 \end{aligned}$$

Since $|x_2 - x_1| = 0.5455 \geq 0.1$, we continue.

Iteration 2

$$\begin{aligned}
x_2 &= 2.4545 \\
f'(2.4545) &= 4(2.4545)^3 - 12(2.4545)^2 + 8(2.4545) \approx 6.4914 \\
f''(2.4545) &= 12(2.4545)^2 - 24(2.4545) + 8 \approx 21.3884 \\
x_3 &= 2.4545 - \frac{6.4914}{21.3884} \approx 2.1510
\end{aligned}$$

Since $|x_3 - x_2| = 0.3035 \geq 0.1$, we continue.

Iteration 3

$$\begin{aligned}
x_3 &= 2.1510 \\
f'(2.1510) &= 4(2.1510)^3 - 12(2.1510)^2 + 8(2.1510) \approx 1.4959 \\
f''(2.1510) &= 12(2.1510)^2 - 24(2.1510) + 8 \approx 11.8989 \\
x_4 &= 2.1510 - \frac{1.4959}{11.8989} \approx 2.0253
\end{aligned}$$

Since $|x_4 - x_3| = 0.1257 \geq 0.1$, we continue.

Iteration 4

$$\begin{aligned}
x_4 &= 2.0253 \\
f'(2.0253) &= 4(2.0253)^3 - 12(2.0253)^2 + 8(2.0253) \approx 0.2104 \\
f''(2.0253) &= 12(2.0253)^2 - 24(2.0253) + 8 \approx 8.6155 \\
x_5 &= 2.0253 - \frac{0.2104}{8.6155} \approx 2.0009
\end{aligned}$$

Since $|x_5 - x_4| = 0.0244 < 0.1$, the algorithm stops. The local minimum is successfully approximated at $x \approx 2.0009$.

Q2. Hooke–Jeeves Method

Solution

Initialization Step Choose a scalar $\varepsilon > 0$ to be used in terminating the algorithm. Choose a starting point $\mathbf{x}_1$, let $\mathbf{y}_1 = \mathbf{x}_1$, let $k = j = 1$, and go to the Main Step.

Main Step

1. Let λ_j be an optimal solution to the problem to minimize $f(\mathbf{y}_j + \lambda \mathbf{d}_j)$ subject to $\lambda \in R$, and let $\mathbf{y}_{j+1} = \mathbf{y}_j + \lambda_j \mathbf{d}_j$. If $j < n$, replace j by $j + 1$, and repeat Step 1. Otherwise, if $j = n$, let $\mathbf{x}_{k+1} = \mathbf{y}_{n+1}$. If $\|\mathbf{x}_{k+1} - \mathbf{x}_k\| < \varepsilon$, stop; otherwise, go to Step 2.

2. Let $\mathbf{d} = \mathbf{x}_{k+1} - \mathbf{x}_k$, and let $\hat{\lambda}$ be an optimal solution to the problem to minimize $f(\mathbf{x}_{k+1} + \lambda \mathbf{d})$ subject to $\lambda \in R$. Let $\mathbf{y}_1 = \mathbf{x}_{k+1} + \hat{\lambda}\mathbf{d}$, let $j = 1$, replace k by $k + 1$, and go to Step 1.

Reference: M. S. Bazaraa, H. D. Sherali, C. M. Shetty, Nonlinear Programming: Theory and Algorithms, 3rd Edition, 2006

$\min f(x_1, x_2) = (x_1 - 2)^4 + (x_1 - 2x_2)^2$

Let $\epsilon = 0.05$. The search directions are:
$\mathbf{d}_1 = (1, 0)$
$\mathbf{d}_2 = (0, 1)$

We start at $\mathbf{x}_1 = (0, 3)$. Note that $\mathbf{x}_1 = (x_1, x_2)$ denotes the input vector in the first iteration, not just the scalar component x_1. Bold vectors are used to prevent confusion! To initialize the method, we set the first temporary base point:
$\mathbf{y}_1 = \mathbf{x}_1 = (0, 3)$.

ITERATION 1

Step 1 ($j = 1$): Find λ_1 minimizing $f(\mathbf{y}_1 + \lambda_1 \mathbf{d}_1) = f(0 + \lambda_1 \cdot 1, 3 + \lambda_1 \cdot 0) = f(\lambda_1, 3)$:

$$\min_{\lambda_1} f(\lambda_1, 3)$$

where $f(\mathbf{y}_1 + \lambda_1 \mathbf{d}_1) = (0 + \lambda_1 - 2)^4 + (0 + \lambda_1 - 6)^2 = (\lambda_1 - 2)^4 + (\lambda_1 - 6)^2$
Based on the problem type you can find an analytical solution, a graphical solution, or apply a 1D line search method (bisection, Newton's method, etc.). Here we show a few steps of Newton's method to solve the 1D line search for λ_1.

$$f'(\mathbf{y}_1 + \lambda_1 \mathbf{d}_1) = 4(\lambda_1 - 2)^3 + 2(\lambda_1 - 6)$$
$$f''(\mathbf{y}_1 + \lambda_1 \mathbf{d}_1) = 12(\lambda_1 - 2)^2 + 2$$

Let $\lambda_1^{(i)}$ denote iteration i of λ_1 in the line search. Let $\lambda_1^{(1)} = 0$ and $\epsilon = 0.001$:

$$\lambda_1^{(2)} = \lambda_1^{(1)} - \frac{f'\left(y_1 + \lambda_1^{(1)} d_1\right)}{f''\left(y_1 + \lambda_1^{(1)} d_1\right)} = 0 - \frac{4(0-2)^3 + 2(0-6)}{12(0-2)^2 + 2} = 0.88 \implies |0.88 - 0| > \epsilon$$

$$\lambda_1^{(3)} = \lambda_1^{(2)} - \frac{f'\left(y_1 + \lambda_1^{(2)} d_1\right)}{f''\left(y_1 + \lambda_1^{(2)} d_1\right)} = 1.81 \implies |1.81 - 0.88| > \epsilon$$

$$\dots$$

$$\lambda_1^{(7)} = \lambda_1^{(6)} - \frac{f'\left(y_1 + \lambda_1^{(6)} d_1\right)}{f''\left(y_1 + \lambda_1^{(6)} d_1\right)} = 3.1352 \implies |3.1352 - 3.2273| > \epsilon$$

$$\lambda_1^{(8)} = \lambda_1^{(7)} - \frac{f'\left(y_1 + \lambda_1^{(7)} d_1\right)}{f''\left(y_1 + \lambda_1^{(7)} d_1\right)} = 3.1282 \implies |3.1282 - 3.1352| > \epsilon$$

$$\lambda_1^{(9)} = \lambda_1^{(8)} - \frac{f'\left(y_1 + \lambda_1^{(8)} d_1\right)}{f''\left(y_1 + \lambda_1^{(8)} d_1\right)} = 3.1282 \implies |3.1282 - 3.1282| \geq \epsilon$$

Hence, $\lambda_1 \approx 3.13$.

$$\mathbf{y}_2 = \mathbf{y}_1 + \lambda_1 \mathbf{d}_1 = (0 + 3.13 \cdot 1, 3) = (3.13, 3)$$

Step 1 ($j = 2$): Since $j = 1 < n = 2$, make $j = 2$ and repeat the process for the next dimension.
Find λ_2 minimizing $f(\mathbf{y}_2 + \lambda_2 \mathbf{d}_2)$ where:

$$f(\mathbf{y}_2 + \lambda_2 \mathbf{d}_2) = (3.13 - 2)^4 + (3.13 - 2 \cdot (3 + \lambda_2))^2 = 1.63 + 4\lambda_2^2 + 11.48\lambda_2 + 9.87$$

Since it is a convex parabola, the minimum with respect to λ_2 can easily be obtained analytically using $-\frac{b}{2a}$:

$$\lambda_2 = -\frac{11.48}{8} = -1.435 \approx -1.44$$

$$\mathbf{y}_3 = \mathbf{y}_2 + \lambda_2 \mathbf{d}_2 = (3.13, 3 - 1.44) = (3.13, 1.56)$$

Check j. Since $j = 2 = n$, the new base point is:

$$\mathbf{x}_2 = \mathbf{y}_3 = (3.13, 1.56)$$

$||\mathbf{x}_2 - \mathbf{x}_1|| = ||(3.13 - 0, 1.56 - 3)|| = \sqrt{3.13^2 + 1.44^2} \geq \epsilon$. Hence, continue.

Step 2: Compute the pattern direction:

$$\mathbf{d} = \mathbf{x}_2 - \mathbf{x}_1 = (3.13 - 0, 1.56 - 3) = (3.13, -1.44)$$

Find $\hat{\lambda}$ minimizing $f(\mathbf{x}_2 + \hat{\lambda}\mathbf{d})$ where

$$\begin{aligned} f(\mathbf{x}_2 + \hat{\lambda}\mathbf{d}) &= f(3.13 + 3.13\hat{\lambda}, 1.56 - 1.44\hat{\lambda}) \\ &= (3.13 + 3.13\hat{\lambda} - 2)^4 + (3.13 + 3.13\hat{\lambda} - 2(1.56 - 1.44\hat{\lambda}))^2 \end{aligned}$$

Again restricting this to a 1D line search, one can find $\hat{\lambda}$ using search methods like Newton's method. Eventually, we converge to $\hat{\lambda} = -0.1$.

$$\mathbf{y}_1 = \mathbf{x}_2 + \hat{\lambda}\mathbf{d} = (3.13 - 0.1 \cdot 3.13, 1.56 - 0.1 \cdot (-1.44)) = (2.82, 1.70)$$

ITERATION 2

Step 1 ($j = 1$): Find λ_1 minimizing $f(\mathbf{y}_1 + \lambda_1\mathbf{d}_1)$, where

$$f(\mathbf{y}_1 + \lambda_1\mathbf{d}_1) = (2.82 + \lambda_1 - 2)^4 + (2.82 + \lambda_1 - 2 \cdot 1.7)^2$$

By Newton's method, $\lambda_1 = -0.12$. Thus,

$$\mathbf{y}_2 = \mathbf{y}_1 + \lambda_1\mathbf{d}_1 = (2.82 - 0.12, 1.7) = (2.70, 1.70)$$

Step 1 ($j = 2$): Check j. Since $j = 1 < n = 2$, repeat the process for $j = 2$.
Find λ_2 minimizing $f(\mathbf{y}_2 + \lambda_2\mathbf{d}_2)$ where

$$f(\mathbf{y}_2 + \lambda_2\mathbf{d}_2) = (2.7 - 2)^4 + (2.7 - 2(1.7 + \lambda_2))^2 = 0.24 + (-0.7 - 2\lambda_2)^2$$

As it is a convex parabola, taking $-\frac{b}{2a}$, we get $\lambda_2 = -\frac{2.8}{8} = -0.35$.

$$\mathbf{y}_3 = \mathbf{y}_2 + \lambda_2\mathbf{d}_2 = (2.7, 1.7 - 0.35) = (2.70, 1.35)$$

Check j. Since $j = 2 = n$, the new base point is:

$$\mathbf{x}_3 = \mathbf{y}_3 = (2.70, 1.35)$$

$||\mathbf{x}_3 - \mathbf{x}_2|| = ||(2.7 - 3.13, 1.35 - 1.56)|| = \sqrt{(-0.43)^2 + (-0.21)^2} = 0.48 > \epsilon$. Hence, continue.

Step 2: Compute the pattern direction:

$$\mathbf{d} = \mathbf{x}_3 - \mathbf{x}_2 = (2.7 - 3.13, 1.35 - 1.56) = (-0.43, -0.21)$$

Find $\hat{\lambda}$ minimizing $f(\mathbf{x}_3 + \hat{\lambda}\mathbf{d})$, where

$$\begin{aligned} f(\mathbf{x}_3 + \hat{\lambda}\mathbf{d}) &= f(2.7 - 0.43\hat{\lambda}, 1.35 - 0.21\hat{\lambda}) \\ &= (2.7 - 0.43\hat{\lambda} - 2)^4 + (2.7 - 0.43\hat{\lambda} - 2(1.35 - 0.21\hat{\lambda}))^2 \end{aligned}$$

Using line search methods, $\hat{\lambda}$ evaluates to 1.5.

$$\mathbf{y}_1 = \mathbf{x}_3 + \hat{\lambda}\mathbf{d} = (2.7 - 1.5 \cdot 0.43, 1.35 - 1.5 \cdot (-0.21)) = (2.06, 1.04)$$

Summary of Iterations

The rest of the iterations can be seen in the following summary table:

Termination: In iteration 5, $||(2 - 2.04, 1 - 1.02)|| = 0.045 < \epsilon$, hence, we stop. The minimum of f is found by $\mathbf{x}^* = (2, 1)$ where $f(2, 1) = (2 - 2)^4 + (2 - 2 \cdot 1)^2 = 0$.

This example can be found in the book (M. S. Bazaraa, H. D. Sherali, C. M. Shetty, Nonlinear Programming: Theory and Algorithms, 3rd Edition, 2006) on page 370.

Iteration k	$\mathbf{x}_k$ $f(\mathbf{x}_k)$	j	$\mathbf{y}_j$	$\mathbf{d}_j$	λ_j	$\mathbf{y}_{j+1}$	$\mathbf{d}$	$\hat{\lambda}$	$\mathbf{y}_3 + \hat{\lambda}\mathbf{d}$
1	(0.00, 3.00)	1	(0.00, 3.00)	(1.0, 0.0)	3.13	(3.13, 3.00)	—	—	—
	52.00	2	(3.13, 3.00)	(0.0, 1.0)	–1.44	(3.13, 1.56)	(3.13, 1.44)	–0.10	(2.82, 1.70)
2	(3.13, 1.56)	1	(2.82, 1.70)	(1.0, 0.0)	–0.12	(2.70, 1.70)	—	—	—
	1.63	2	(2.70, 1.70)	(0.0, 1.0)	–0.35	(2.70, 1.35)	(–0.43, –0.21)	1.50	(2.06, 1.04)
3	(2.70, 1.35)	1	(2.06, 1.04)	(1.0, 0.0)	–0.02	(2.04, 1.04)	—	—	—
	0.24	2	(2.04, 1.04)	(0.0, 1.0)	–0.02	(2.04, 1.02)	(–0.66, –0.33)	0.06	(2.00, 1.00)
4	(2.04, 1.02)	1	(2.00, 1.00)	(1.0, 0.0)	0.00	(2.00, 1.00)	—	—	—
	0.000003	2	(2.00, 1.00)	(0.0, 1.0)	0.00	(2.00, 1.00)			
5	(2.00, 1.00)								
	0.00								

Reference: M. S. Bazaraa, H. D. Sherali, C. M. Shetty, Nonlinear Programming: Theory and Algorithms, 3rd Edition, 2006

Q3. Steepest Descent

Solution

Summary of the Steepest Descent Algorithm

Given a point $\mathbf{x}$, the steepest descent algorithm proceeds by performing a line search along the direction $-\nabla f(\mathbf{x})/\|\nabla f(\mathbf{x})\|$ or, equivalently, along the direction $-\nabla f(\mathbf{x})$. A summary of the method is given below.

Initialization Step Let $\varepsilon > 0$ be the termination scalar. Choose a starting point $\mathbf{x}_1$, let $k = 1$, and go to the Main Step.

Main Step

If $\|\nabla f(\mathbf{x}_k)\| < \varepsilon$, stop; otherwise, let $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$, and let λ_k be an optimal solution to the problem to minimize $f(\mathbf{x}_k + \lambda \mathbf{d}_k)$ subject to $\lambda \geq 0$. Let $\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$, replace k by $k+1$, and repeat the Main Step.

Reference: M. S. Bazaraa, H. D. Sherali, C. M. Shetty, Nonlinear Programming: Theory and Algorithms, 3rd Edition, 2006

Initialization

First, we find the gradient vector $\nabla f(\mathbf{x})$ by calculating the partial derivatives with respect to x_1 and x_2:

$$\frac{\partial f}{\partial x_1} = e^{x_1} - e^{-x_1}$$
$$\frac{\partial f}{\partial x_2} = e^{x_2} - e^{-x_2}$$

Thus, the gradient vector is:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} e^{x_1} - e^{-x_1} \\ e^{x_2} - e^{-x_2} \end{bmatrix}$$

The method of steepest descent updates the current point using the rule:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k$$

where $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$ is the search direction, and λ_k is the optimal step size found via a numerical line search to minimize $f(\mathbf{x}_k + \lambda \mathbf{d}_k)$ subject to $\lambda \geq 0$. We start with initial point $\mathbf{x}_1 = (4, 3)$ and termination scalar $\epsilon = 0.01$.

Iteration 1 ($k = 1$)

1. Evaluate Function at $\mathbf{x}_1 = (4.0000, 3.0000)$:

$$\begin{aligned} f(\mathbf{x}_1) &= e^{4.0000} + e^{-4.0000} + e^{3.0000} + e^{-3.0000} \\ &= 54.5982 + 0.0183 + 20.0855 + 0.0498 \\ &= 74.7518 \end{aligned}$$

2. Evaluate Gradient and Search Direction:

$$\nabla f(\mathbf{x}_1) = \begin{bmatrix} e^{4.0000} - e^{-4.0000} \\ e^{3.0000} - e^{-3.0000} \end{bmatrix} = \begin{bmatrix} 54.5982 - 0.0183 \\ 20.0855 - 0.0498 \end{bmatrix} = \begin{bmatrix} 54.5799 \\ 20.0357 \end{bmatrix}$$

$$\mathbf{d}_1 = -\nabla f(\mathbf{x}_1) = \begin{bmatrix} -54.5799 \\ -20.0357 \end{bmatrix}$$

3. Calculate Gradient Norm:

$$\begin{aligned} \|\nabla f(\mathbf{x}_1)\| &= \sqrt{(54.5799)^2 + (20.0357)^2} \\ &= \sqrt{2978.9655 + 401.4293} = \sqrt{3380.3948} = 58.1412 \end{aligned}$$

Since $\|\nabla f(\mathbf{x}_1)\| = 58.1412 > 0.01$, we proceed.

4. Line Search for λ_1 and Update Point: We perform a line search to minimize $f(\mathbf{x}_1 + \lambda \mathbf{d}_1)$ subject to $\lambda \geq 0$. Solving this numerically yields the optimal step size $\lambda_1 \approx 0.0842$.

$$\begin{aligned} \mathbf{x}_2 &= \begin{bmatrix} 4.0000 \\ 3.0000 \end{bmatrix} + 0.0842 \begin{bmatrix} -54.5799 \\ -20.0357 \end{bmatrix} \\ &= \begin{bmatrix} 4.0000 - 4.5956 \\ 3.0000 - 1.6870 \end{bmatrix} = \begin{bmatrix} -0.5956 \\ 1.3130 \end{bmatrix} \end{aligned}$$

Iteration 2 ($k = 2$)

1. Evaluate Function at $\mathbf{x}_2 = (-0.5956, 1.3130)$:

$$\begin{aligned} f(\mathbf{x}_2) &= e^{-0.5956} + e^{0.5956} + e^{1.3130} + e^{-1.3130} \\ &= 0.5512 + 1.8141 + 3.7173 + 0.2690 \\ &= 6.3516 \end{aligned}$$

2. Evaluate Gradient and Search Direction:

$$\nabla f(\mathbf{x}_2) = \begin{bmatrix} e^{-0.5956} - e^{0.5956} \\ e^{1.3130} - e^{-1.3130} \end{bmatrix} = \begin{bmatrix} 0.5512 - 1.8141 \\ 3.7173 - 0.2690 \end{bmatrix} = \begin{bmatrix} -1.2629 \\ 3.4483 \end{bmatrix}$$

$$\mathbf{d}_2 = -\nabla f(\mathbf{x}_2) = \begin{bmatrix} 1.2629 \\ -3.4483 \end{bmatrix}$$

3. Calculate Gradient Norm:

$$\begin{aligned} \|\nabla f(\mathbf{x}_2)\| &= \sqrt{(-1.2629)^2 + (3.4483)^2} \\ &= \sqrt{1.5949 + 11.8908} = \sqrt{13.4857} = 3.6723 \end{aligned}$$

Since $\|\nabla f(\mathbf{x}_2)\| = 3.6723 > 0.01$, we proceed.

4. Line Search for λ_2 and Update Point: We minimize $f(\mathbf{x}_2 + \lambda \mathbf{d}_2)$ numerically, yielding the optimal step length $\lambda_2 \approx 0.3915$.

$$\begin{aligned} \mathbf{x}_3 &= \begin{bmatrix} -0.5956 \\ 1.3130 \end{bmatrix} + 0.3915 \begin{bmatrix} 1.2629 \\ -3.4483 \end{bmatrix} \\ &= \begin{bmatrix} -0.5956 + 0.4944 \\ 1.3130 - 1.3500 \end{bmatrix} = \begin{bmatrix} -0.1012 \\ -0.0370 \end{bmatrix} \end{aligned}$$

Summary Table of Computations

The iterative process continues similarly. The summary of computations for 4 iterations is provided below. All values are rounded to four decimal places.

Iteration k	$\mathbf{x}_k$ $f(\mathbf{x}_k)$	$\nabla f(\mathbf{x}_k)$	$\lVert\nabla f(\mathbf{x}_k)\rVert$	$\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$	λ_k	$\mathbf{x}_{k+1}$
1	$(4.0000, 3.0000)$ 74.7518	$(54.5799, 20.0357)$	58.1412	$(-54.5799, -20.0357)$	0.0842	$(-0.5956, 1.3130)$
2	$(-0.5956, 1.3130)$ 6.3516	$(-1.2629, 3.4483)$	3.6723	$(1.2629, -3.4483)$	0.3915	$(-0.1012, -0.0370)$
3	$(-0.1012, -0.0370)$ 4.0116	$(-0.2027, -0.0740)$	0.2158	$(0.2027, 0.0740)$	0.4993	$(0.0000, -0.0001)$
4	$(0.0000, -0.0001)$ 4.0000	$(0.0000, -0.0002)$	0.0002	—	—	—

Termination Step:

At the beginning of Iteration 4, the gradient is evaluated as $\lVert\nabla f(\mathbf{x}_4)\rVert = 0.0002$. Because this value is strictly less than the selected termination scalar $\epsilon = 0.01$, the algorithm stops. The global minimum is successfully found near $\mathbf{x}^* \approx (0, 0)$ with a function value of approximately 4.0000.